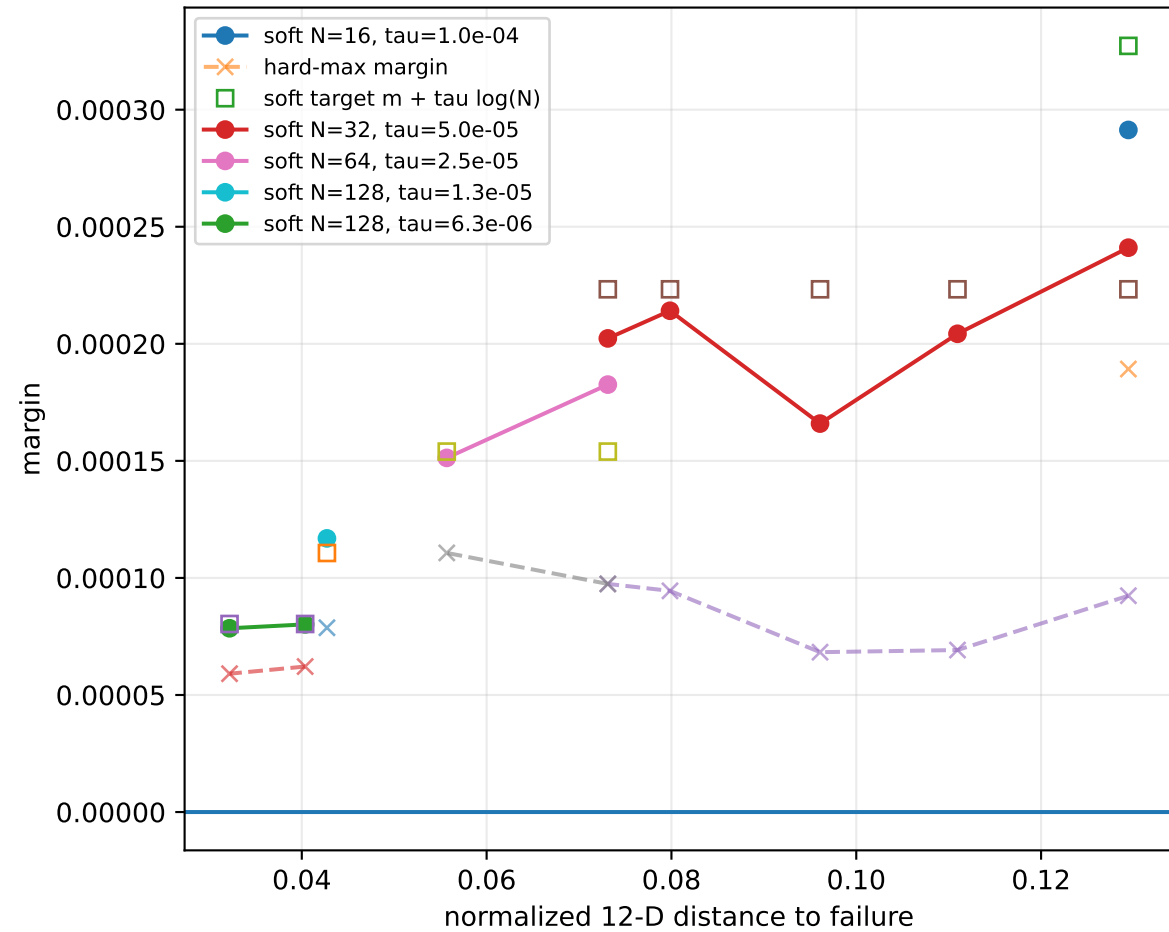
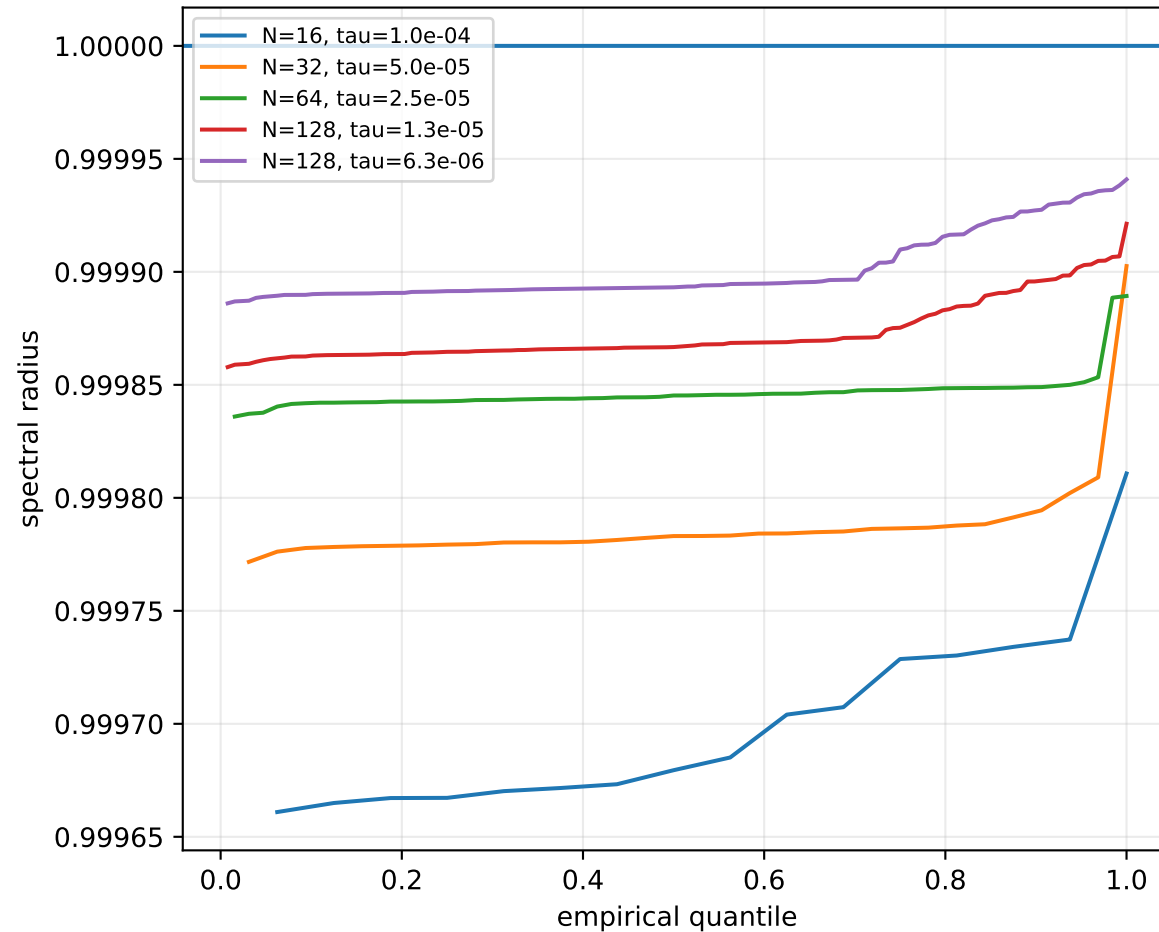


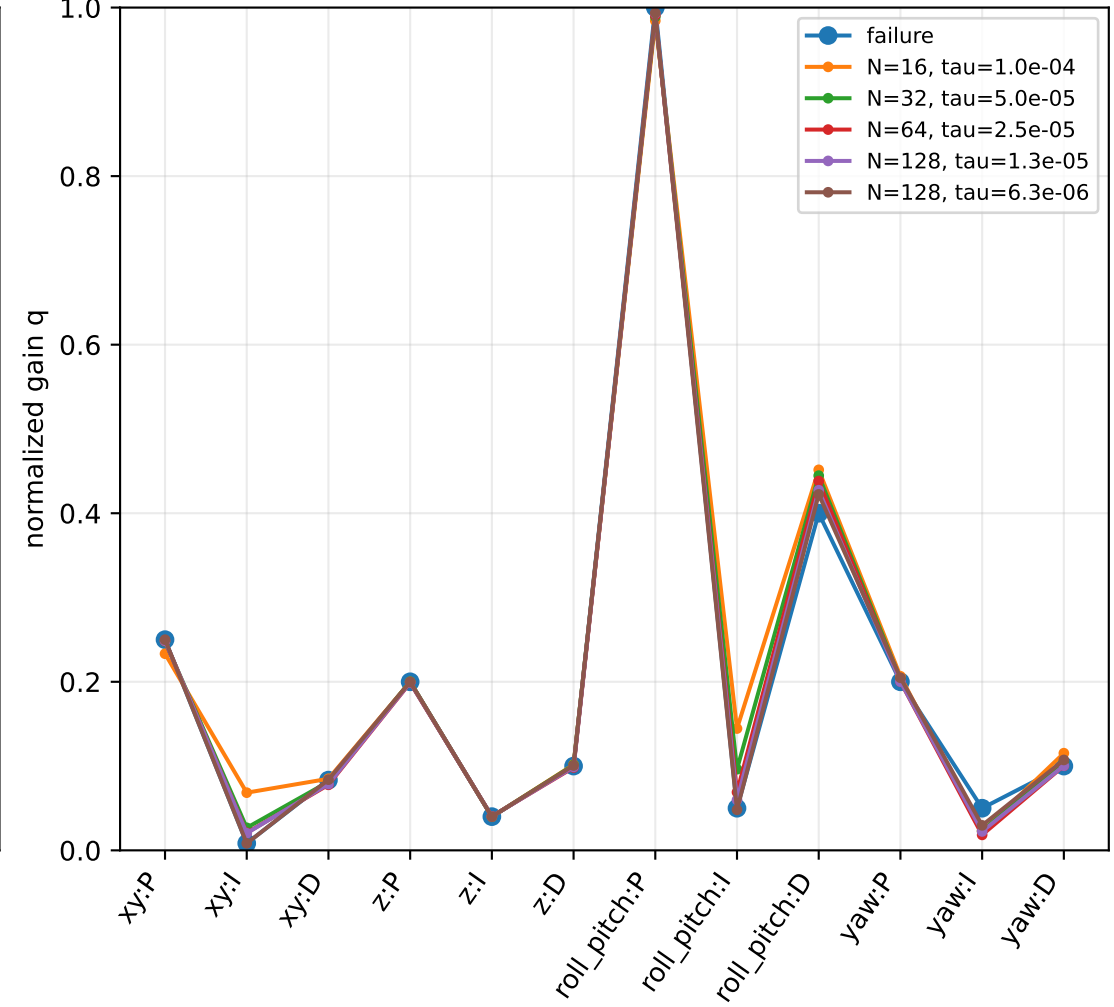
Target-level continuation and sharpening



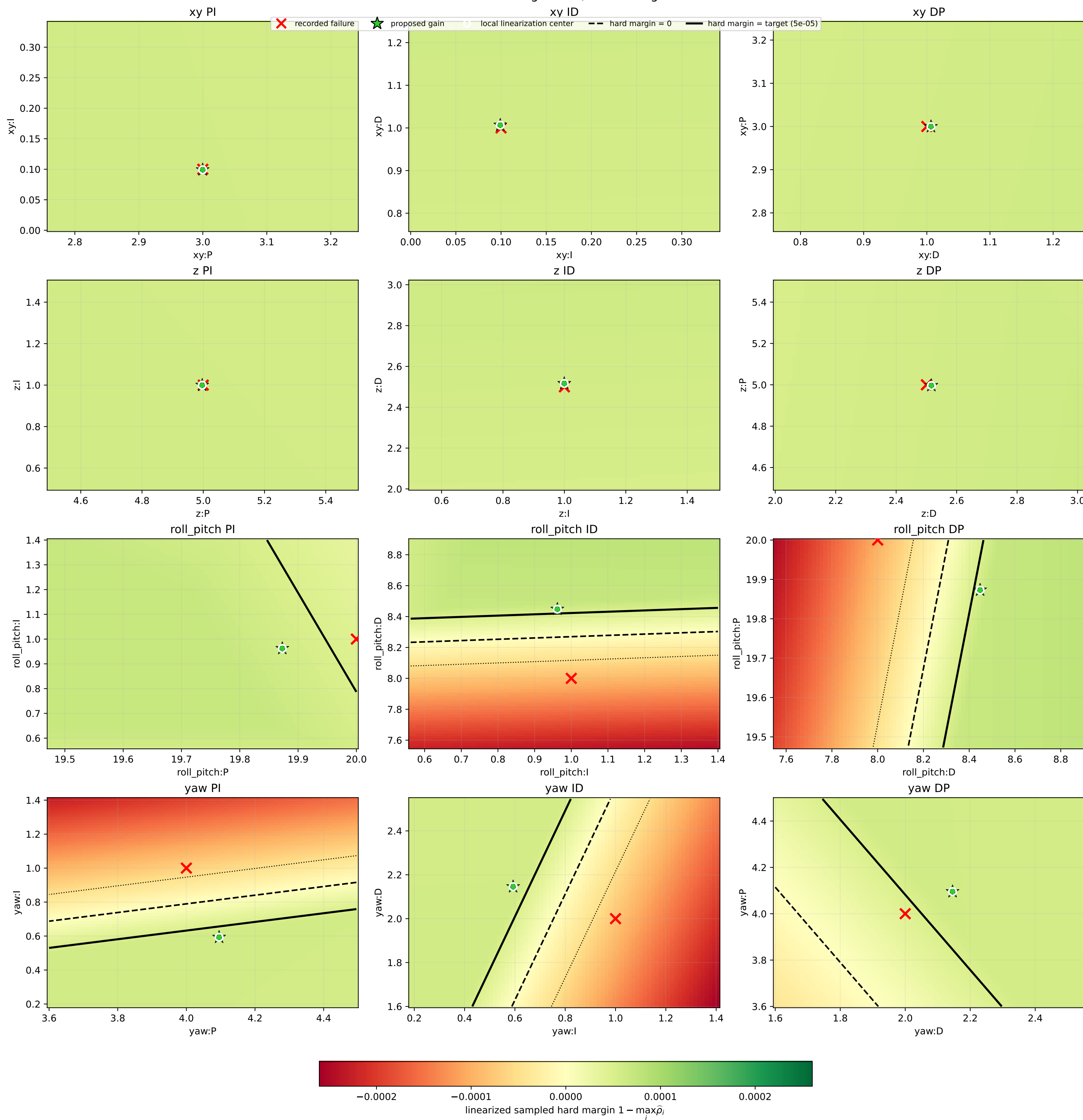
Order-statistic curve at each sharpened target level



PID displacement from failure



Linearized local sampled hard-margin field (green = safer, red = unsafe)
Contours are local first-order diagnostics, not exact global boundaries



PID gain update: recorded failure to hard-margin proposal

Group	Gain	Failure	Proposed	Delta	Delta / cap
xy	P	3	2.999761	-0.0002391649	-1.993041e-05
xy	I	0.1	0.09912841	-0.0008715911	-7.263259e-05
xy	D	1	1.006534	+0.006534312	+0.000544526
z	P	5	4.996537	-0.003462746	-0.0001385098
z	I	1	0.9995041	-0.000495874	-1.983496e-05
z	D	2.5	2.516449	+0.01644876	+0.0006579502
roll_pitch	P	20	19.87295	-0.1270521	-0.006352604
roll_pitch	I	1	0.9619925	-0.03800747	-0.001900373
roll_pitch	D	8	8.447574	+0.4475743	+0.02237872
yaw	P	4	4.09536	+0.09535987	+0.004767993
yaw	I	1	0.5928657	-0.4071343	-0.02035672
yaw	D	2	2.145638	+0.1456385	+0.007281924

Hard-margin target m: 5e-05
 Soft target at final tau: 8.032519e-05
 Final sample count N_max: 128
 Nested Gaussian-QMC prefixes: 16 -> 32 -> 64 -> 128
 Initial / final tau: 0.0001 / 6.25e-06
 Exact final soft margin: +7.849562e-05
 Exact final hard margin: +5.909111e-05
 Final soft-level residual: -1.830e-06
 Normalized 12-D distance from failure: 0.03218205

The proposal is evaluated with the exact sampled forward model. Contour fields on page 2 are local first-order diagnostics only.